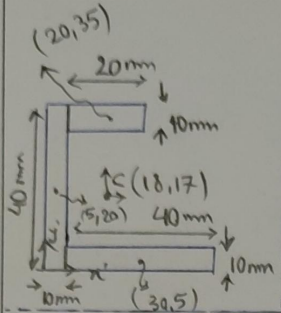


مقرر سہ سو سو درجہ مکمل سارے یسوفہ .

نور پورہ حقیران - 404201984

سوال اول: پرسی تسی حلل ازخمش



$$\bar{x}_c = \frac{\sum A_i \bar{x}_i}{\sum A_i} = \frac{(40 \times 10)(5) + (40 \times 10)(30) + (20 \times 10)(20)}{40 \times 10 + 40 \times 10 + 20 \times 10} = 18 \text{ mm}$$

$$\bar{y}_c = \frac{\sum A_i \bar{y}_i}{\sum A_i} = \frac{(40 \times 10)(20) + (40 \times 10)(5) + (20 \times 10)(35)}{40 \times 10 + 40 \times 10 + 20 \times 10} = 17 \text{ mm}$$

$$I_{xx} = \sum \left(\frac{1}{12} b h^3 + A \Delta y^2 \right), \quad I_{yy} = \sum \left(\frac{1}{12} h b^3 + A \Delta x^2 \right), \quad I_{xy} = \sum A \Delta x \Delta y, \quad (\bar{x}_c, \bar{y}_c) = (18, 17) \text{ mm}$$

$$I_{xx} = \frac{1}{12} (40^3 \times 10) + (40 \times 10)(20-17)^2 + \frac{1}{12} (10^3 \times 20) + (10 \times 20)(35-17)^2 + \frac{1}{12} (10^3 \times 40) + (10 \times 40)(5-17)^2$$

$$I_{yy} = \frac{1}{12} (40 \times 10^3) + (40 \times 10)(5-18)^2 + \frac{1}{12} (10 \times 20^3) + (10 \times 20)(20-18)^2 + \frac{1}{12} (10 \times 40^3) + (10 \times 40)(30-18)^2$$

$$I_{xy} = (10 \times 40)(5-18)(20-17) + (20 \times 10)(20-18)(35-17) + (40 \times 10)(30-18)(5-17)$$

$$I_{xx} = \frac{1}{3} (553000) \text{ mm}^4, \quad I_{yy} = 227600 \text{ mm}^4, \quad I_{xy} = -66000 \text{ mm}^4$$

$P_x = P \sin 10^\circ, \quad P_y = P \cos 10^\circ \Rightarrow M_x = +P(h-z) \sin 10^\circ, \quad M_y = -P(h-z) \sin 10^\circ$

$u = h - z$

$$\tan \alpha = \frac{M_y I_{xx} - M_x I_{xy}}{M_x I_{yy} - M_y I_{xy}} = \frac{-P \sin 10^\circ \times \frac{553000}{3} - P \cos 10^\circ \times (-66000)}{-P \cos 10^\circ \times 227600 + P \sin 10^\circ \times (-66000)} = -0.4117$$

$$\alpha = \tan^{-1}(-0.4117) = -22.378^\circ, \quad (\bar{x}_c, \bar{y}_c) = (18, 17) \text{ mm}, \quad \alpha = -22.378^\circ$$

$$\text{rot: } (z=0, \quad \sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$$\sigma_{z=h} = 0.89x + 7.07y \Rightarrow \sigma_{z=h_A} = 0.89(0-18) + 7.07(40-17) = 146.59 \text{ N/mm}^2 \text{ (tension)}$$

$$\sigma_{z=h_B} = 0.89(40-18) + 7.07(0-17) = -100.61 \text{ N/mm}^2$$

$$\sigma_{h_{\max}} = 146.59 \text{ N/mm}^2, \quad \sigma_{h_{\min}} = -100.61 \text{ N/mm}^2$$

$$\text{Displacement: } u'' = \frac{M_x I_{xy} - M_y I_{xx}}{I = E(I_{xx} I_{yy} - I_{xy}^2)}, \quad u'(z=0) = 0, \quad u(z=0) = 0$$

$$v'' = \frac{+M_x I_{yy} + M_y I_{xy}}{I = E(I_{xx} I_{yy} - I_{xy}^2)}, \quad v'(z=0) = 0, \quad v(z=0) = 0$$

2. حساب إجهادات القص في المثلث.

404201984 - (مراجعة)

أول سؤال أول:

$$u'' = \frac{1}{I} (M_x I_{xy} - M_y I_{xx}) = \frac{1}{I} (P \cos 10^\circ (h-z) + P \sin 10^\circ (h-z)) = \frac{P(I_{xy} \cos 10^\circ + I_{xx} \sin 10^\circ)}{I} (h-z)$$

$$u' = \frac{P}{I} (\cos 10^\circ + \sin 10^\circ) (hz - \frac{z^2}{2} + A) \quad , \quad u = \frac{P}{I} (I_{xy} \cos 10^\circ + I_{xx} \sin 10^\circ) (\frac{hz^2}{2} - \frac{z^3}{6} + Az + B)$$

$$\text{From B.C}^* : A=B=0 \Rightarrow u = \frac{P(I_{xy} \cos 10^\circ + I_{xx} \sin 10^\circ)}{E(I_{xx} I_{yy} - I_{xy}^2)} (\frac{1}{2} h z^2 - \frac{1}{6} z^3) \quad \times (h-z)$$

$$v'' = \frac{1}{I} (-M_x I_{xy} + M_y I_{yy}) = \frac{1}{I} (-P(h-z) I_{xy} \cos 10^\circ - P(h-z) I_{xy} \sin 10^\circ) = \frac{-P(I_{xy} \cos 10^\circ + I_{xy} \sin 10^\circ)}{I}$$

$$v' = \frac{-P}{I} (I_{xy} \cos 10^\circ + I_{xy} \sin 10^\circ) (hz - \frac{z^2}{2} + A') \quad , \quad v = \frac{-P}{I} (I_{xy} \cos 10^\circ + I_{xy} \sin 10^\circ) (\frac{hz^2}{2} - \frac{z^3}{6} + A'z + B')$$

$$\text{From B.C}^{**} : A'=B'=0 \Rightarrow v = \frac{-P(I_{xy} \cos 10^\circ + I_{xy} \sin 10^\circ)}{E(I_{xx} I_{yy} - I_{xy}^2)} (\frac{1}{2} h z^2 - \frac{z^3}{6})$$

$$Z = 500 \text{ mm}, h = 1000 \text{ mm} \Rightarrow u_{Z=500 \text{ mm}} = -1.587 \text{ mm}, v_{Z=500 \text{ mm}} = -10.229 \text{ mm}$$

$$E = 72 \text{ GPa} = 72000 \text{ N/mm}^2$$

$$(\bar{x}, \bar{y}) = (18.17) \text{ mm}, \alpha = -22.378^\circ \quad \underline{\text{الف}}$$

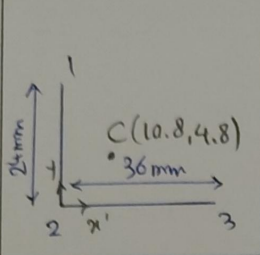
$$\sigma_{\max_{Z=0}} = 146.59 \text{ N/mm}^2, \sigma_{\min_{Z=0}} = -100.61 \text{ N/mm}^2 \quad \underline{\text{ب}}$$

$$u_{Z=500} = -1.587 \text{ mm}, v_{Z=500} = -10.229 \text{ mm} \quad \underline{\text{ج}}$$

• دینا سہی سونہ کون قائل سارن پیرسرفہ •

• 404201984 - نورب یوریاتہیان •

اموال دوم : نس غشی وایرہاکی سزی



$$\bar{x}_c = \frac{\sum x_i A_i}{\sum A_i} = \frac{18 \times 36 \times 1.25}{24 \times 1.25 + 36 \times 1.25} = 10.8 \text{ mm}$$

$$\bar{y}_c = \frac{\sum y_i A_i}{\sum A_i} = \frac{12 \times 24 \times 1.25}{24 \times 1.25 + 36 \times 1.25} = 4.8 \text{ mm}$$

$$I_{xx} = \frac{1}{12} 24^3 \times 1.25 + 24 \times 1.25 \times (12 - 4.8)^2 + 36 \times 1.25 \times 4.8^2 = 4032 \text{ mm}^4$$

$$I_{yy} = \frac{1}{12} 36^3 \times 1.25 + 36 \times 1.25 (18 - 10.8)^2 + 24 \times 1.25 (10.8)^2 = 10692 \text{ mm}^4$$

$$I_{xy} = 24 \times 1.25 (0 - 10.8)(12 - 4.8) + 36 \times 1.25 (18 - 10.8)(0 - 4.8) = -3888 \text{ mm}^4$$

tip : $M_y = 80(z) = 0$, $M_x = 50(z) = 0$ $\Rightarrow M_y = M_x = 0$

middle : $M_y = 80 \times 100 = 8000 \text{ Nmm}$, $M_x = -50 \times 100 = -5000 \text{ Nmm}$

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

$\sigma_{z \text{ tip}} = 0$, $\sigma_{z \text{ mid}} = 0.46x - 0.80y$

for $\underline{z}$: $\sigma_{z \text{ tip}} = 0$, $\sigma_{z \text{ mid}} = 0.46(0 - 10.8) - 0.80(0 - 4.8) = -1.128 \text{ N/mm}^2$

$I_{xx} = 4032 \text{ mm}^4$, $I_{yy} = 10692 \text{ mm}^4$, $I_{xy} = -3888 \text{ mm}^4$ الب

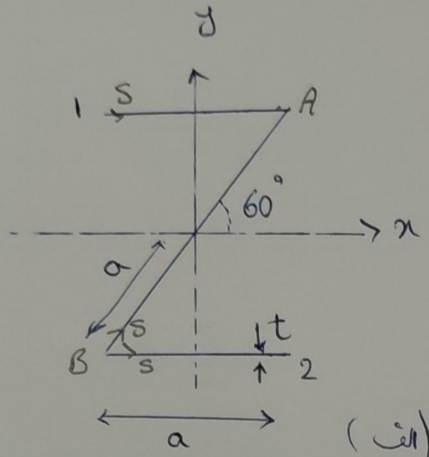
$\sigma_{z \text{ tip}} = 0 \text{ N/mm}^2$, $\sigma_{z \text{ mid}} = 0.46x - 0.80y$ ج

$\sigma_{z \text{ t}} = -0.876 \text{ N/mm}^2$, $\sigma_{z \text{ m}} = -1.128 \text{ N/mm}^2$ ج

مقررہ سی سوں دریں تحلیل سارہ یسیرت .

404201984- (رپورقیقیان)

سؤال سوم : محاسبه مساحت در مقطع متقاطع .



$$I_{xx} = \int y^2 dA = \int y^2 t ds = t \int y^2 ds$$

$$y \text{ for } 2B: y = -a \sin 60^\circ \text{ for } 0 < s_1 < a$$

$$y \text{ for } 1A: y = a \sin 60^\circ \text{ for } 0 < s_2 < a$$

$$y \text{ for } AB: y = s \sin 60^\circ - a \sin 60^\circ \text{ for } 0 < s_3 < 2a$$

$$I_{xx} = t \int y^2 ds$$

$$I_{xx} = t \left[\int_0^a (a \sin 60^\circ)^2 ds_1 + \int_0^a (s \sin 60^\circ - a \sin 60^\circ)^2 ds_3 \right]$$

$$= 2t (a^2 \sin^2 60^\circ) a + t \int_0^{2a} (s \sin 60^\circ - a \sin 60^\circ)^2 ds_3$$

$$= 2t \left(\frac{3}{4} a^3 \right) + \frac{(2a)^3}{4} + \frac{3}{4} 2a^3 - (2a) a \left(\frac{3}{4} \right) = \frac{3}{2} ta^3 + \frac{1}{2} ta^3 = 2ta^3$$

$$I_{yy} = \int x^2 dA = t \int x^2 ds, \quad x \text{ for } 1A: x = s_4 - a/2 \text{ for } 0 < s_4 < a; \quad 2B \text{ same}$$

$$x \text{ for } AB: x = s_5 \cos 60^\circ - a \cos 60^\circ \text{ for } 0 < s_5 < 2a$$

$$I_{yy} = t \left[\int_0^a (s_4 - a/2)^2 ds_4 + \int_0^{2a} (s_5 \cos 60^\circ - a \cos 60^\circ)^2 ds_5 \right] = t \left[\int_0^a (s_4^2 - a s_4 + a^2/4) ds_4 + \int_0^{2a} (s_5^2 \cos^2 60^\circ - a s_5 \cos 60^\circ + a^2 \cos^2 60^\circ) ds_5 \right]$$

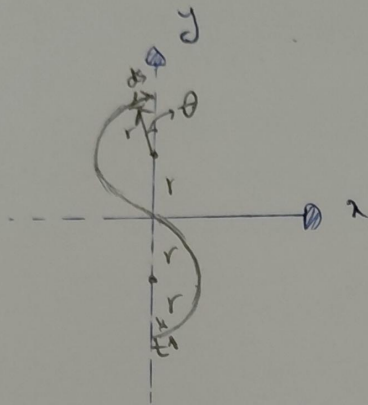
$$I_{xy} = \int xy dA = t \int xy ds = t \int (a \sin 60^\circ)(s_1 - a/2) ds_1 + t \int (-a \sin 60^\circ)(s_2 - a/2) ds_2$$

$$+ t \int (s_3 \sin 60^\circ - a \sin 60^\circ)(s_3 \cos 60^\circ - a \cos 60^\circ) ds_3 = t \int (s_3 \sin 60^\circ \cos 60^\circ - a^2 \sin 60^\circ \cos 60^\circ - 2a s_3 \sin 60^\circ \cos 60^\circ + a^2 \sin^2 60^\circ) ds_3$$

$$= \sqrt{3}/4 t \left(\frac{s_3^3}{3} + a^2 s_3 - a s_3^2 \right) \Big|_0^{2a} = \sqrt{3}/4 t \left(\frac{2}{3} a^3 \right) = \sqrt{3}/6 ta^3$$

$$I_{xx} = 2ta^3, \quad I_{yy} = \frac{1}{3} ta^3, \quad I_{xy} = \sqrt{3}/6 ta^3$$

ادامہ سوال سوم :



$$\begin{aligned} I_{xx} &= 2 \int y^2 dA = 2 \int y^2 ds = 2t \int_0^\pi (r + r \cos \theta)^2 d\theta \\ &= 2tr^3 \int_0^\pi (1 + \cos \theta)^2 d\theta = 2tr^3 \int_0^\pi (1 + 2\cos \theta + \cos^2 \theta) d\theta \\ &= 2tr^3 \left(\theta + 2\sin \theta + \frac{\theta}{2} + \frac{1}{4}\sin 2\theta \right) \Big|_0^\pi = 3\pi tr^3 \end{aligned}$$

$$(\rightarrow) I_{yy} = 2 \int x^2 dA = 2 \int (\sin \theta r)^2 tr d\theta = 2tr^3 \int \sin^2 \theta d\theta$$

$$2tr^3 \int \frac{1}{2}(1 + \cos 2\theta) d\theta = 2tr^3 \left(\frac{\theta}{2} + \frac{1}{4}\sin 2\theta \right) \Big|_0^\pi = \pi tr^3$$

$$I_{xy} = 2 \int xy dA = 2 \int_0^\pi (r + r \cos \theta)(-r \sin \theta) tr d\theta = -2tr^3 \int_0^\pi (\sin \theta + \sin \theta \cos \theta) d\theta$$

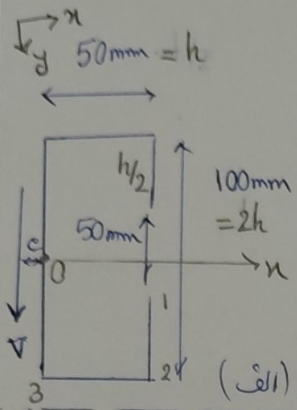
$$= -2tr^3 \left((-\cos \theta - \frac{1}{4}\cos 2\theta) \Big|_0^\pi \right) = -4\pi tr^3$$

$$I_{xx} = 3\pi tr^3, I_{yy} = \pi tr^3, I_{xy} = -4\pi tr^3$$

404201984 - روبرت پورچفیل

تاریخ سرکاری (میں نے اسے تسلیم کیا ہے)

مسئلہ چھٹا: مرکز پرت (c)



$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t x ds - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int_0^s t y ds$$

$$S_x = 0, I_{xy} = 0 \text{ (چونکہ } \omega \text{)}, t = \text{constant}$$

$$\Rightarrow q_s = - \frac{S_y}{I_{xx}} \int t y ds, k = 50 \text{ mm}$$

$$I_{xx} = \frac{1}{12} t (2h)^3 + (2)(kt)(h^2) + 2 \frac{1}{12} t (h/2)^3 + (2)(t)(h/2)(3h/4)^2 = \frac{39}{12} t h^3 *$$

$$q_s = - \frac{v t}{I_{xx}} \int y ds$$

$$1-2: y = -s_1 - h/2, 0 \leq s_1 \leq h/2 \Rightarrow q_{s1} = \frac{-vt}{I_{xx}} \int_0^{s_1} -(s_1 + h/2) ds_1 = \frac{vt}{I_{xx}} s_1^2, q_2 = \frac{vh^2}{4I_{xx}}$$

$$2-3: y = -h, 0 \leq s_2 \leq h \Rightarrow q_{s2} = \frac{-vt}{I_{xx}} \int_0^{s_2} (-h) ds_2 + q_2 = \frac{vt}{I_{xx}} s_2 + \frac{vh^2}{4I_{xx}}, q_3 = \frac{5vt}{4I_{xx}}$$

$$\sum M_o = 0 \Rightarrow Vc = \int q p ds = \int_0^{h/2} \frac{2vt}{I_{xx}} s_1^2 (h) ds_1 + \int_0^h \left(\frac{2vt}{I_{xx}} s_2 + \frac{vh^2}{4I_{xx}} \right) (h/2) ds_2$$

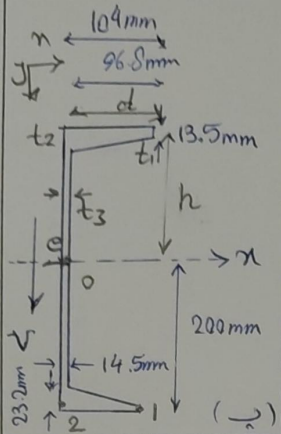
$$Vc = \frac{2vt}{I_{xx}} \frac{s_1^3}{3} \Big|_0^{h/2} + (h/2) \left(\frac{2vt}{I_{xx}} \frac{s_2^2}{2} + \frac{vh^2}{4I_{xx}} s_2 \right) \Big|_0^h = \frac{5}{6} \frac{vh^4}{I_{xx}} \Rightarrow c = \frac{24}{39} h$$

$$c = 30.77 \text{ mm}$$

• اورینٹ یونیورسٹی - 404701984

• تھری سٹرکچرل سٹینڈرڈ ڈیزائن سائز پینل

ادامہ سوال چھٹا :



$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t_x dx - \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t_y dy$$

$$S_x = 0, I_{xy} = 0 \text{ (مقنن)} \Rightarrow q_s = - \frac{S_y}{I_{xx}} \int t_y dy$$

$$I_{xx} = \frac{1}{12} t_3 (2h)^3 + (2) \left(\frac{t_1 + t_2}{2} \right) (d) (h^2) = \frac{h^2}{3} (2t_3 h + 3(t_1 + t_2) d)$$

$$1-2: y = -h, 0 \leq s_1 \leq d, t = t_1 + \left(\frac{t_2 - t_1}{d} \right) s_1$$

$$q_{12} = \frac{-S_y}{I_{xx}} \int_0^{s_1} t_y dy = \frac{-S_y}{I_{xx}} \int_0^{s_1} \left[t_1 + \left(\frac{t_2 - t_1}{d} \right) s_1 \right] (-h) ds_1 = \frac{S_y h}{I_{xx}} \left[t_1 s_1 + \left(\frac{t_2 - t_1}{d} \right) \frac{s_1^2}{2} \right]$$

$$\sum M_o = 0 \Rightarrow V_e = \int p q ds = 2 \int_0^d \frac{V h}{I_{xx}} \left[t_1 s_1 + \left(\frac{t_2 - t_1}{d} \right) \frac{s_1^2}{2} \right] h ds$$

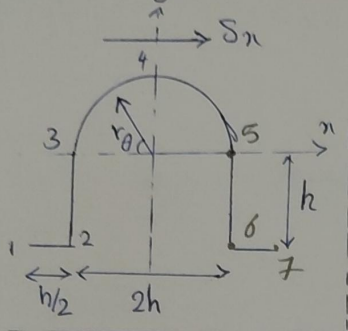
$$V_e = \frac{2 V h^2}{I_{xx}} \left(\frac{t_1 s_1^2}{2} + \left(\frac{t_2 - t_1}{d} \right) \frac{s_1^3}{6} \right) \Big|_0^d = \frac{V h d^2}{3 I_{xx}} (2t_1 + t_2) \Rightarrow e = \frac{d^2 (2t_1 + t_2)}{2t_3 h + 3(t_1 + t_2) d}$$

$$e = 28.58 \text{ mm}$$

• مرکز ثقل درین شکل سازه سیرفت.

• 404201984 - (پروژه مقدماتی)

سوال پنجم: توزیع جریان میانی



$$q_s = - \left(\frac{S_x I_{xx} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t r ds + \left(\frac{S_y I_{yy} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t y ds$$

$$S_y = 0, I_{xy} = 0 \text{ (مختار)}, h = r = 50 \text{ mm} \Rightarrow q_s = \frac{-S_x}{I_{yy}} \int t r ds$$

$$I_{yy} = \left[\frac{1}{12} + (h/2)^3 + (h/2)(5/4h)^2 + (h)(h^2) + \int_0^{\pi/2} (r \cos \theta)^2 r d\theta \right] 2 = \left(\frac{43 + 6\pi}{12} \right) h^3$$

$$1-2: x = -\frac{3}{2}h + s_1, 0 \leq s_1 \leq h/2 \Rightarrow q_{12} = \frac{-S_x}{I_{yy}} \int_0^{s_1} \left(-\frac{3}{2}h + s_1 \right) ds = \frac{-S_x}{2I_{yy}} (s_1^2 - 3hs_1), q_2 = \frac{5S_x h^2}{8I_{yy}}$$

$$2-3: x = -h, 0 \leq s_2 \leq h \Rightarrow q_{23} = \frac{-S_x}{I_{yy}} \int_0^{s_2} (-h) ds_2 + \frac{5}{8} \frac{S_x h^2}{I_{yy}} = \frac{-S_x}{I_{yy}} \left(hs_2 + \frac{5}{8}h^2 \right), q_3 = \frac{13}{8} \frac{S_x h^2}{I_{yy}}$$

$$3-4: x = -r \cos \theta, 0 \leq \theta \leq \pi/2 \Rightarrow q_{34} = \frac{-S_x}{I_{yy}} \int_0^\theta -r \cos \theta r d\theta + \frac{13}{8} \frac{S_x h^2}{I_{yy}} = \frac{h^2 S_x}{I_{yy}} \left(\sin \theta + \frac{13}{8} \right)$$

مقدار q در نقاط مختلف نسبت به خط معیار، نسبت است و معادلات هم‌انگشتی، بدون در نظر گرفتن مقدار اولیه، می‌توانند معادله‌ای برای q در هر یک از دست می‌آید.

$$q_4 = \frac{21}{8} \frac{h^2}{I_{yy}} S_x$$

$$q_1 = 0 = q_7$$

$$0 \leq s_1 \leq h/2, \quad q_{12} = \frac{-S_x}{I_{yy}} \left(\frac{3}{2}hs_1 - \frac{s_1^2}{2} \right), \quad q_2 = q_6 = \frac{-S_x}{I_{yy}} \left(\frac{5}{8}h^2 \right)$$

$$0 \leq s_2 \leq h, \quad q_{23} = \frac{-S_x}{I_{yy}} \left(hs_2 + \frac{5}{8}h^2 \right), \quad q_3 = q_5 = \frac{-S_x}{I_{yy}} \left(\frac{13}{8}h^2 \right)$$

$$0 \leq \theta \leq \pi/2, \quad q_{34} = \frac{-S_x}{I_{yy}} \left(\sin \theta h + \frac{13}{8}h^2 \right), \quad q_4 = \frac{-S_x}{I_{yy}} \left(\frac{21}{8}h^2 \right)$$

$$0 \leq \theta \leq \pi/2, \quad q_{45} = \frac{-S_x}{I_{yy}} \left(\frac{21}{8}h^2 - \sin \theta h^2 \right)$$

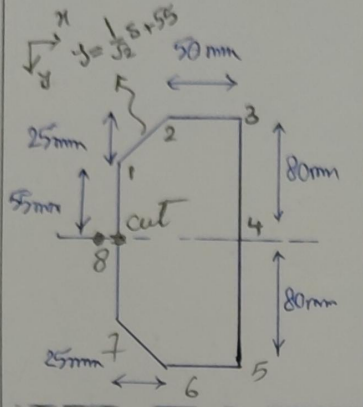
$$0 \leq s_3 \leq h, \quad q_{56} = \frac{-S_x}{I_{yy}} \left(\frac{13}{8}h^2 - hs_3 \right)$$

$$0 \leq s_4 \leq h/2, \quad q_{67} = \frac{-S_x}{I_{yy}} \left(\frac{5}{8}h^2 - \frac{3}{2}hs_4 + \frac{s_4^2}{2} \right)$$

• قطر اسفل سوراخ 50 میلی باشد مرکز آن در مرکز باشد.

• 404701984 - پروفیسر

مسئله: مقطع بیست



cut:

$$q_b = - \left(\frac{S_y I_{xx} - S_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t_{xds} - \left(\frac{S_x I_{yy} - S_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) \int t_y ds$$

$$S_x = 0, I_{xy} = 0 \text{ (نکته)} \Rightarrow q_b = \frac{-S_y}{I_{xx}} \int t_y ds$$

$$I_{xx} = \frac{1}{12} (110)^3 + \frac{1}{12} 160^3 + (50 \times 80)^2 + \int \left(\frac{1}{\sqrt{2}} s + 55 \right)^2 ds$$

$$= \frac{1}{12} (110)^3 + \frac{1}{12} 160^3 + (50 \times 80)^2 + 4 \left(\frac{(25\sqrt{2})^3}{6} + 55^2 \times 25 \times \sqrt{2} + \frac{55}{\sqrt{2}} 25^2 \right) = 5 \times 10^6 \text{ mm}^4$$

$$\text{for } 01: y = s_1, 0 \leq s_1 \leq 55 \Rightarrow q_{b1} = \frac{-S_y t}{I_{xx}} \int_0^{s_1} s_1 ds = \frac{-S_y t}{I_{xx}} \frac{s_1^2}{2}, q_{b1} = \frac{-S_y}{I_{xx}} \frac{3025}{2}$$

$$\text{for } 12: y = \frac{1}{\sqrt{2}} s_2 + 55, 0 \leq s_2 \leq 25\sqrt{2} \Rightarrow q_{b2} = \frac{-S_y t}{I_{xx}} \int_0^{s_2} \left(\frac{1}{\sqrt{2}} s_2 + 55 \right) ds + \frac{-S_y}{I_{xx}} \frac{3025}{2}$$

$$= \frac{-S_y t}{I_{xx}} \left(\frac{1}{\sqrt{2}} \frac{s_2^2}{2} + 55 s_2 + \frac{3025}{2} \right), q_{b2} = \frac{-S_y t}{I_{xx}} (3898.98)$$

$$\text{for } 23: y = 80, 0 \leq s_3 \leq 50 \Rightarrow q_{b3} = \frac{-S_y t}{I_{xx}} \int_0^{s_3} 80 ds - \frac{S_y t}{I_{xx}} (3898.98) = \frac{-S_y t}{I_{xx}} (80 s_3 + 3898.98)$$

$$q_{b3} = \frac{-S_y t}{I_{xx}} (7898.98)$$

$$\text{for } 34: y = 80 - s_4, 0 \leq s_4 \leq 80 \Rightarrow q_{b4} = \frac{-S_y t}{I_{xx}} \int_0^{s_4} (80 - s_4) ds - \frac{S_y t}{I_{xx}} (7898.98)$$

$$q_{b4} = \frac{-S_y t}{I_{xx}} \left(80 s_4 - \frac{s_4^2}{2} + 7898.98 \right), q_{b4} = \frac{-S_y t}{I_{xx}} (11098.98)$$

$$\text{shear loads path through shear center: } q_{s,0} = \frac{-\oint q_b ds}{\oint ds} \quad \text{Gt = cte} \quad \oint ds = \sum s_i = 370 + 50\sqrt{2} \text{ mm}$$

$$-\oint q_b ds = \frac{-2S_y t}{I_{xx}} \left[\int_0^{55} \frac{s_1^2}{2} ds_1 + \int_0^{25\sqrt{2}} \left(\frac{\sqrt{2}}{4} s_2^2 + 55 s_2 + 1512.5 \right) ds_2 + \int_0^{50} (80 s_3 + 3898.98) ds_3 \right.$$

$$\left. + \int_0^{80} \left(80 s_4 - \frac{s_4^2}{2} + 7898.98 \right) ds_4 \right] = \frac{S_y t^2}{I_{xx}} (1173909.017) \Rightarrow q_{s,0} = \frac{S_y t^2}{I_{xx}} (5327.35)$$

پیشہ سرکاری سروس میں ملحقہ سارے پیشہ

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اداریہ سوال نمبر 3

$$q_{01} = q_{b01} + q_0 = \frac{-s_y t}{I_{xx}} \left(\frac{1}{2} s_1^2 - 5327.35 \right), 0 \leq s_1 \leq 55$$

$$q_{12} = q_{b12} + q_0 = \frac{-s_y t}{I_{xx}} \left(\frac{\sqrt{2}}{4} s_2^2 + 55 s_2 - 3814.85 \right), 0 \leq s_2 \leq 25\sqrt{2}$$

$$q_{23} = q_{b23} + q_0 = \frac{-s_y t}{I_{xx}} \left(80 s_3 - 1428.37 \right), 0 \leq s_3 \leq 50$$

$$q_{34} = q_{b34} + q_0 = \frac{-s_y t}{I_{xx}} \left(80 s_4 - \frac{1}{2} s_4^2 + 2571.63 \right), 0 \leq s_4 \leq 80$$

$$q_i' = q_3 = q_{34} s_{=0} = \frac{-s_y t}{I_{xx}} (2571.63), q_{ii}' = q_1 = q_{12} s_{=0} = \frac{s_y t}{I_{xx}} (3814.85)$$

$$I_{xx} = 5 \times 10^6 \text{ mm}^4, t = 4 \text{ mm} \Rightarrow q_i' = -2.057 \times 10^{-3} s_y, q_{ii}' = 3.0518 \times 10^{-3} s_y$$

$$I_{xx} = 5 \times 10^6 \text{ mm}^4, t = 4 \text{ mm}$$

$$q_{01} = \frac{-s_y t}{I_{xx}} \left(0.5 s_1^2 - 5327.35 \right), 0 \leq s_1 \leq 55$$

$$q_{12} = \frac{-s_y t}{I_{xx}} \left(0.35 s_2^2 + 55 s_2 - 3814.85 \right), 0 \leq s_2 \leq 25\sqrt{2}$$

$$q_{23} = \frac{-s_y t}{I_{xx}} \left(80 s_3 - 1428.37 \right), 0 \leq s_3 \leq 50$$

$$q_{34} = \frac{-s_y t}{I_{xx}} \left(80 s_4 - 0.5 s_4^2 + 2571.63 \right), 0 \leq s_4 \leq 80$$

$$q_i' = -2.057 \times 10^{-3} s_y, q_{ii}' = 3.0518 \times 10^{-3} s_y$$

$$q_{45} = \frac{-s_y t}{I_{xx}} \left(5771.63 - 80 s_5 + \frac{1}{2} s_5^2 \right), 0 \leq s_5 \leq 80, q_{56} = \frac{-s_y t}{I_{xx}} \left(2571.63 - 80 s_6 \right), 0 \leq s_6 \leq 50$$

$$q_{67} = \frac{-s_y t}{I_{xx}} \left(-1428.37 - \frac{\sqrt{2}}{4} s_7^2 - 55 s_7 \right), 0 \leq s_7 \leq 25\sqrt{2}, q_{78} = \frac{-s_y t}{I_{xx}} \left(3814.85 - \frac{1}{2} s_8^2 \right), 0 \leq s_8 \leq 55$$